Periodic Bottle Beams

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1 Introduction

This contains equations to generate a periodic bottle beam.

2 Math

Consider Fourier transform of a phase mask $\phi(r,\theta) = n\theta$:

$$\int_0^R \int_0^{\Pi} e^{i\phi(r,\theta)} e^{-ikr\cos(\theta-\phi)} r dr d\theta = 2\pi i n J_n(kr) \tag{1}$$

where J_n is the Bessel function of the first kind

3 quadratic bottle beam profile

$$A_{bb}(k,z) = \int_0^a r\, \exp\Big[-r^2\Big(\tfrac{1}{w_{\text{in}}^2}+i\,\tfrac{z}{2f}\Big)\Big]\, J_0(kr)\, dr - \int_a^H r\, \exp\Big[-r^2\Big(\tfrac{1}{w_{\text{in}}^2}+i\,\tfrac{z}{2f}\Big)\Big]\, J_0(kr)\, dr$$

$$A_{bb}(0,z) = \int_0^a r\, \exp\Big[-r^2\Big(\tfrac{1}{w_{\text{in}}^2}+i\,\tfrac{z}{2f}\Big)\Big]\, dr \;-\; \int_a^H r\, \exp\Big[-r^2\Big(\tfrac{1}{w_{\text{in}}^2}+i\,\tfrac{z}{2f}\Big)\Big]\, dr$$

Let $\alpha = \dfrac{1}{w_{\text{in}}^2}+i\,\dfrac{z}{2f}$. Then $A_{bb}(0,z) = \displaystyle\int_0^a r\, e^{-\alpha r^2}\, dr - \int_a^H r\, e^{-\alpha r^2}\, dr = \frac{1-2e^{-\alpha a^2}+e^{-\alpha H^2}}{2\alpha}$.

$$A_{bb}(0,z) \;=\; \frac{w_{\text{in}}^2}{2}\Big(1-2e^{-a^2/w_{\text{in}}^2}+e^{-H^2/w_{\text{in}}^2}\Big) + \frac{i\,z}{4f}\Big[w_{\text{in}}^2\big(2a^2e^{-a^2/w_{\text{in}}^2}-H^2e^{-H^2/w_{\text{in}}^2}\big)-w_{\text{in}}^4\big(1-2e^{-a^2/w_{\text{in}}^2}+e^{-H}$$

$$A_{bb}(0,z) \;=\; \frac{1-2\,e^{-\alpha a^2}}{2\alpha} \quad \text{with} \quad \alpha = \frac{1}{w_{\text{in}}^2} + i\,\frac{z}{2f}.$$

$$A_{bb}(0,z) \;=\; \frac{1-2\exp\!\Big(-\big(\frac{1}{w_{\text{in}}^2}+i\,\frac{z}{2f}\big)\,w_{\text{in}}^2\ln 2\Big)}{2\Big(\frac{1}{w_{\text{in}}^2}+i\,\frac{z}{2f}\Big)}.$$

$$A_{bb}(0,z) = \frac{1-\exp\!\Big(-i\,\frac{w_{\text{in}}^2\ln 2}{2f}\,z\Big)}{\dfrac{2}{w_{\text{in}}^2}+i\,\dfrac{z}{f}}.$$

$$I_{bb}(0,z) = \big|A_{bb}(0,z)\big|^2 = \frac{\Big|1-\exp\!\big(-i\,\frac{w_{\text{in}}^2\ln 2}{2f}\,z\big)\Big|^2}{\Big|\frac{2}{w_{\text{in}}^2}+i\,\frac{z}{f}\Big|^2}.$$

$$I_{bb}(0,z) = \frac{\Big|1-e^{-i\frac{w_{\text{in}}^2\ln 2}{2f}z}\Big|^2}{\Big(\frac{2}{w_{\text{in}}^2}\Big)^2+\Big(\frac{z}{f}\Big)^2} = \frac{2-2\cos\!\big(\frac{w_{\text{in}}^2\ln 2}{2f}z\big)}{\dfrac{4}{w_{\text{in}}^4}+\dfrac{z^2}{f^2}} = \frac{4\sin^2\!\big(\frac{w_{\text{in}}^2\ln 2}{4f}z\big)}{\dfrac{4}{w_{\text{in}}^4}+\dfrac{z^2}{f^2}}.$$

$$I_{bb}(0,z) = \frac{w_{\text{in}}^8(\ln 2)^2}{16\,f^2}\,z^2 - \frac{w_{\text{in}}^{12}(\ln 2)^2\big((\ln 2)^2+12\big)}{768\,f^4}\,z^4 + O(z^6)\,.$$

4 Figure

